

ActInf MathStream 007.1 ~ "Constructor Theory as Process Theory", Vincent Wang-Mascianica et al.

Source: [YouTube](#) Playlist: MathStreams Duration: 1:36:38 | Uploaded:

Unknown Transcript method: auto_caption

foreign hello and welcome everyone it is September 19th 2020-23 and we are in octave math stream number 7.1 at the Active inference Institute we're here with Vincent Wang mashinita and guests it should be quite a discussion certainly a unique one by what we can already see so thank you all for joining we'll be looking forward to this presentation and interleave in discussion thanks Dan um so hi my name is Vincent and uh with me today also is Hamza in the audience uh we're both authors on this paper that we'll be talking today Constructor Theory as process Theory um so uh well to the format of this discussion be I'd like to tell you a bit about constructive theory and process Theory and then sort of have a discussion about uh the the sort of diagrammatics of process Theory how that might fit with active inference um and all this you know you can view it as an exercise and sort of seeing things in a diagrammatic way um so I'll get started and the structure will be roughly I'll talk about what Constructor theory is first just from a conceptual level because I'm the only non-physicist among the authors of this paper hands as a physicist so any questions about specific examples is there to be directed to Hamza uh and then you know after that I'm going to do just a sort of lightning fast introduction to what a process theory is and then we'll put the two together and then we'll have a discussion and throughout I will be making pauses to sort of invite questions and make sure everyone is on the same page conceptually but everything should be super super easy because I think all the ideas really are you know fundamentally they're quite simple Okay so without further ado what is Constructor Theory instructor NASA so you may have heard of this already because there's quite a lot of hype they have a they have a very good PR machine if nothing else so among other things that's been described as and I think this is from quanta article a master Theory a set of ideas so fundamental that all other theories would spring from it and in fact if you go to deutsche's original on constructive theory he goes so far to say that all other theories of physics you know we'll call them you know it doesn't matter if it's thermodynamics or Quantum or relativity gravity whatever we'll just call them subsidiary

theories because that's how big and important constructive theory is and uh you know there's this eponymous book by Kiara Maletto the science of Canon cats I want to say that oh well you know if constructive theory is about counterfactuals and what what is and isn't possible in the domain of physics um and all this is all this sounds really impressive but you know the question is what do you mean by what do you mean by this so how this is how this all started is uh some of Kiara's students Anisette and uh Maria Villares they came over to the Wolfson Quantum foundations discussion and they were nice enough to sit down and tell us about what Constructor Theory really is because you know historically it's been really difficult to sort of pin down you know to pin down like what is what do you really mean here by all these words and uh well these you know they're happy to say look you know it's a meta Theory you can sort of bring your own mathematics you know what we really there's a sort of methodological approach of uh constructive theory that's more important than the specific mathematics that you want to use to instantiate all these words like what is a Constructor and what is possible and what is impossible but okay I'll tell you conceptually what constructive theory is so first I want I want you to let's talk about a space of conceivable processes conceivable is the word here the use of the word conceivable will become clear soon because um we're gonna we're gonna cut this space shortly and what are the sorts of conceivable processes that we might have well let's let's look at a limited subset let's talk about ice cream a sort of the sort of simple process that can always happen which is you might have some ice cream you leave it up for too long and the ice cream melts comes a puddle this is you know we want to say this is obviously a conceivable process there's another conceivable process where the opposite thing happens you have the ice cream that's melted and then it spontaneously unmelts and becomes an ice cream again this is a conceivable process as well and then there's a straight up you know ridiculous stuff that you know we we know for sure is not going to happen in the domain it says it's like for example we have an ice cream and then you've got some machine such that you can take it and you can take you put in anything and it performs it performs this process or task of taking your input and copying it everything perfectly down to the down to the quantum States you know you're you're cloning the ice cream and you may be aware of this Quantum no cloning so we know that this this sort of thing you know not going to happen nevertheless all of these are conceivable processes um of relationships between possible inputs and possible outputs I should stop using the word possible just inputs and outputs right so what is the aim of Constructor Theory really well it stems from this desire to classify the spaces of conceivable processes into that which is possible and that which is impossible well

what might we say intuitively we might say that well we we're pretty sure that we're pretty sure that this this ice cream melting stuff that's possible in terms of some Theory of physics that explains everything like okay fine like some Theory physics that explains what happens to ice cream we definitely want to say that ice cream melting is possible um and we definitely you know that Theory of physics also has got to say that this uh sort of cloning situation here that's got to be that's got to be no good that's that's impossible one of the difficulty comes when you've got the cases in the Middle where okay well technically if we're just using English I mean it is conceivable it is possible that an ice cream can spontaneously uh reform itself I mean it can happen the laws of physics don't forbid it it is simply very very unlikely um what is deutsche's idea that we should say no no no you know let's do away with the complexities of statistics and just say that that's straight up impossible right so then the goal of constructive theory is to take the space of conceivable processes and cut it just one cut corresponding to on one side what is impossible and on the other what is possible right so just using this one sort of concept or Duality what is possible and impossible it is like you take your space of conceivable processes and you just determine what's possible and impossible and here's the subtle part then you claim that any Theory of physics corresponds precisely to the content of one of these cuts and what do I mean by that well I mean is that well you know here's this is one space of conceivable processes that happens to talk about for example the Dynamics of ice cream but there are all kinds of conceivable spaces of processes that might govern you know different sort of domain or domains or or magisteria or whatever you want to call them of a phenomena and the idea is that well you know anytime you have a theory of physics what do you really have you really have just a domain of phenomena that you care about and Constructor Theory says and in addition to that a cut that tells you whether you know what things are possible in that domain and what things are not possible and so then every pair of you know sort of domain Plus one of these cuts that tells you what is possible and impossible is going to kind of encode or summarize just be a particular Theory of physics and so we can be very empathetic to this this uh we can be very empathetic to this kind of formulation it's a nice thing to try to do to sort of uh to summarize or have a sort of bird's eye view of all these different uh theories of physics to try to boil it down to something that's conceptually a bit simple but of course there are problems too right so one problem is you know why didn't you call it a possibility Theory rather than constructive theory I don't know the answer to that but possibility Theory sounds a bit too much like a mathematical Theory and it doesn't sound exactly as constructive theory so that's one reason you might want to call

it a Constructor Theory but then what is a Constructor what do Constructors have to do with all of this this idea of possibility and impossibility well you see the difficulty was if you go back and you think about the case of the ice cream melting that's something that is possible by our usage of the word but you'd want to be able to stick it on one side of the cut and call that the solution unfortunately is herbicide what you do is you simply make up a technical concept you call this concept the possible you call this concept possibility of a task let's say and then you well what do you mean by possibility or task well that relies on you know defining what these Constructors are the possibility defines depends on defining what Constructors are and Constructors in turn depend on on the definition of tasks and then tasks finally are something that you can cash out in terms of processes so this is this is the idea of where constructive theory wants to go and now I'm going to tell you how it is that Constructor theorists get there uh let's see do we have do we have any questions at this point any sort of conceptual questions or clarifications that anyone wants to because otherwise I'm gonna move on where does likely or unlikely fit that's all on one side of the line what if something's vanishingly unlikely but possible that is an absolutely brilliant question um and the answer is you get to choose as long as you make a cut because so the dogma of constructive theory goes you only get possible and impossible I don't care how you pick you pick how to distinguish the likely eventually unlikely and I didn't make up the rules I'm merely telling you that everything has to be either possible or impossible good question sorry that's the best that's the best I can do okay three so what is Constructor Theory all right it's a of course it's a meta Theory bring your own math the medicine what is that what does that mean that means bring your own math okay let's start from what is a construct you know what this might be this might be better if I you know I'm looking at I'm looking at these notes now and I'm going like nope I'm gonna it wouldn't make sense if I started from there I'm gonna have to put that as the as the bottom thing and then we'll get there what is uh instructor right okay let me start with defining what a task is so all of this so far is going to be in English and then we're going to translate to math in a second but hold on let's let's talk about what a task is first from a task you get onto this remember we we said that something was conceivable we have to break down that initial diagram that we drew this initial informal thing we we had the sort of the space of all conceivable task so first we have to Define what do we mean by tasks what do we mean by conceivable tasks and then from concealable let's just talk about what do we mean by a possible task right and then it will turn out that a Constructor is a the Constructor is uh is something that's associated with a possible task so that you see that there's a difficulty here of getting from Constructors to the you know

where ultimately you want to go there's a sort of yeah right logic of justification logic of Discovery and they don't know differently okay fine so what is a task well remember we're allowed to we're allowed to bring our own mathematics here so we we asked them we we took a lot of care to ask them you know what what sort of math do you want you know what do you need here and there and then finally you know it came down to look look really what we care about as Constructor theorists is just defining these things once you pick some sort of sensible interpretation of these things in mathematics then you can go ahead and play the game of Constructor Theory so okay that's fantastic you know so so here we have a here in Oxford we have a uh we have a nice sort of Homebrew math what we you know we call them process theories but otherwise they're known as symmetrical categories let me say okay well let's let's call let's call a task a process a process in a symmetric monoidal category okay well we'll get to what that means in a second so what is a conceivable task that becomes well that becomes mathematically trivial it just becomes a process inside a process Theory well it is a possible task well possible task is going to be and then here's where we have to get technical again what is the possible task well it's defined as some as an infinitely iterable task infinitely iterable yada yada okay and then if a task turns out to be one of these infinitely if a task turns out to be infinitely iterable we call it possible and a task always is doing something and then there's something in the environment that's helping it do it that's something in the environment this super ancillary thing at the end of these definitions is going to be what a Constructor is and at this point it will be very helpful to actually go through an example and it so happens that by going through this example just in the diagrammatic form you will understand process theories because the whole point of process theories and symmetrical categories is that you have this formal diagrammatic syntax that's extremely intuitive you don't have to know anything about the category Theory you can just use the diagrams and there's a formal syntax where you sort of kept all wrong as long as you're just like composing the boxes and the wires so I'm going to now give an example let's make an example of a constructible and this is also a good time so now that I'm erasing anytime I'm racing it's a good time to chime in with questions sure one question from the chat upcycle Club wrote does this mean that Constructor Theory as a process Theory indeed takes into account the node deleting theorem I think um no no deleting what do you what do you mean I'm not familiar with not sure if that's what was written or maybe you referenced no cloning earlier I represents no cloning ah so here's the yeah so here's the nice bit so earlier I drew a diagram where there are all these different Blobs of different colors and each one was a different uh domain of physics

or whatever right so it turns out that you know part of the content of a process Theory or the symmetric model category is that you have to pick a category an ambient category in which you're interpreting and that choice of ambient category uh tells you about the sorts that that is the choice of the conceivable processes so if you choose the category of quantum processes and Quantum Maps it will turn out that in that setting you have no cloning but if you pick a different setting like for example maybe you want just sets and functions you know mathematical sets and functions in which case of course you can copy you can always you know copy down any information that you have you know as long as all on paper and you're just using sets and functions right but in physics find you we want to model it with the process theory of uh of quantum Maps then you don't get you know so it depends on it depends on the choice of uh choice of interpreting category but that's that's a very good conceptual question thank you talk about a task I'm going to make a task here I'm going to draw these tasks as directed boxes with wires um so this is this is just one of my drawings here okay I've drawn these directed so because I want everything to be read from left to right now in a process Theory every process is drawn as a box the process is something that takes as input over here some systems and does something to it whatever then it outputs some systems and then processes can be composed in you know a kind of usual way that one expects this is a composite process for example all right what is the process so what is the process here well okay let's let's talk about uh these systems what I mean by these systems can be just about anything it was subject to a couple of rules the rules are anytime you have a system you just need some way to talk about what it means to have two of these systems side by side so let's say that the green system is uh it's either the green system here so now this is something that takes in a green system and an orange system and gives back the green system of the orange system so it's something that's acting on these systems that returns them and what I mean by a system or a system can be it will mathematically we would call them a type like a system can be like the booleans or they they can be a set or they can even have the vector spaces as systems and if you have Vector spaces as systems there are lots of different ways that you can interpret this idea of sticking to systems side by side you can have the direct sum of vector spaces you have the tensor product and Vector spaces and all of them will give you valid process theories so you have linear maps with direct sum or linear maps with tensor products and that would all give you a process Theory or asymmetrical category in which you can interpret the diagrams you're drawing here so let's let's cut with a simple one though a simple one is let's say that this green system is going to be you know this is terribly drawn oh that's a

slightly better so it's shoes all right it's a shoe let's say it's a particular issue right that helps and what is this other system so what is this other system I'm going to call this other system the system of a bucket with a paintbrush and Chuck it and here's a paintbrush this is perfectly fine systems talk about these systems can be in different states like for example a shoe can be well it can be clean or it can be dirty it can have all these different colors with the bucket well let's talk about buckets you know sort of having paint inside maybe the bucket has some paint in that maybe it doesn't that's a good example of the system and I want to model a particular task here which is Paint The Shoe right it's a process well another why do I call it a task well to turn up to to view a process as a task I have to I have to look at it in the right way I have to say like okay well there's got to be one there's got to be one sort of that's the input and output I want to call that system the substrate well why because the task has to you know that that's the substrate there and you you know painting has to happen to something if you want to view painting as a task in the substrate of the painting is going to be the shoe and what about this other what about the what about the paint well the paint part okay well that's going to be part of what we call just everything else that's going to be the environment is going to be a substrate an but then what do we what do we mean by successfully painting a shoe well okay so here I'm going to need to introduce we need to induce a concept so green wire is the shoe system as this orange wire is the bucket and paint system let's remember that so now here's the new kind of process maybe yeah yeah I think about this like this this is fine here's a new kind process so I'm drawing Foster season you've seen the process at this point that takes inputs and has outputs what about a process that doesn't take any inputs and just gives you an output well those we call States the special kind of process in the process theory of sets and functions or symmetrical model category I should say of sets and functions with Cartesian products states are elements of sets in the process theory of relations states are subsets of sets and systems are sets in uh in I think any vect whether whether it's a direct sum or the tensor product a state is a vector so States really are just sort of like some you're really we really call it you know it's a sort of play on words because I was saying like oh well you know the system can have States their systems can be in different states the idea is that well actually the name is well chosen so that a state really is picking out you know one of the states of the system that you're interested in let's say that we'll call there's there's a sort of state of the shoe which is that it's unpainted and there's a state of the bucket and bucket and paint system such that you got a bucket that's full of blue paint and what do I mean by the task of painting the shoe let's see how much is visible oh I've got all of that all of the side over here right now you'll see your first process

theoretic equation so here's a shoe that's not blue and here's a bucket of paint that's now it's got no more blue paint in it so what we got here let's say here so here's full right here let's say it's let's say it's white starts off the white shoe at the ends up the painted I have here empty and this is a process theoretic equation I hope you can see all the data that's happening here what we're saying is that okay look if I've got as initial States I've got like a I've got an unpainted shoe and I've got a full bucket of paint what this process does is it takes the two systems it views one of them as a it just keeps track and says okay well this one's a substrate this one's the what is the result of applying that process to my two states well then I have to end up with two new States and those states are saying well okay now I end up with a painted shoe and I use up all my paint so now I haven't I have an empty can of paint That's not mistake is that clear because that's that's if you understand this then you'll understand everything else I'm about to say this is maybe the only this is the only conceptually difficult part on constructive theory as process Theory are we happy with this if anyone's unhappy or has any conceptual questions and now's the right now's the right time to ask it seems like we kind of put a box around the entire Crux of the issue but at this level of abstraction indeed right indeed right I mean like yeah of course you uh there's a lot of there's a lot of boxing away the Crux of the issue but in fact we haven't even gotten to rather we haven't even gotten to the the Crux of what the Constructor is yet I'm merely defining what it is for something to be a task I mean you can then pick so so the point also of a process theory is that you can work at the level of abstraction you so choose think of this you know if you if you code this is basically you're declaring what the types of the the method or the function are it's still on you to go fill in the implementation details but here we're just dealing with the level of this this is the level of abstraction where it's it's perfectly okay to start working to just Define what we mean by a Constructor and remember that this is still just an example at some point you can just erase all the labels and just keep the coloring that indicates the types and then that would this would be the abstract specification of what a task in a Constructor is yeah so you're right we are hiding away a lot of the or hiding away you know the Crux of the difficulty or the meat but we're doing so in service of sort of getting at a formal incarnation of what's constructive theory is so yeah um Okay so kind of happy with this now I'm going to redo some of this let's see so now well let's suppose that it's you know maybe it's a small bucket of pain it's a fairly big shoe in either case the content of the equation was that you know you stick in this full bucket of paint and then at the end of it you run out of paint you finish painting issue but you run out of paint what that means as a sort of diagrammatic consequence what happens if we want to paint a

shoe twice well I mean by that okay well here 's two shoes and just to distinguish them I mean uh just distinguish them let's say that this one starts off like a nice yellow shoe and this one's a nice Orange Shoe and then in the end we still have this we have this again this bucket of paint that is kind of that's full what do I mean by painting painting Two Shoes well okay let's let's paint them in the order that they're closest so first we want to apply the the paint shoe task right we end up with some shoe system we end up with some bucket of paint system we want to say okay we'll send that that first system we send it away that second system we bring it in all right now let's line things up again and let's apply this again here's the paint shoe and here's you know let's try it again let's try to paint the shoe right and now we've we're seeing a new diagrammatic component which corresponds to the word symmetric in a symmetrical category the idea is that well in a symmetrical model category yes you can put systems side by side but in a sense you don't care about the spatial configuration of how you do it you're always free it's you sort of have a bag of systems you know to pick them out and geometrically what this amounts to in the diagrammatic syntax is saying that well look if you have a system side by side you can always you can always swap the two systems and you can depict the swaps as uh as wires twisting over each other and the sort of the the difficulty of uh or the only difficult content of the symmetrical category is showing that you know the the way that the diagrams behave topologically correspond precisely to the algebraic content of of the the morphisms that are the semantics of the of the processes which means which which is to say that as long as you're tracking the connectivity and the connectivity matches between two different diagrams they're going to refer to the same formal entity uh so there is so it's really like a safe syntax to to use right but we can follow this right this is the this is this is what I said I'm painting I'm trying to paint the two shoes with one bucket of paint that should be clear I'm trying to arrange to paint the first to the orange shoe and then I twist in the second yellow shoe and try to try to apply that process again to do the painting right and then remember the diagrammatic equations the diagrammatic equations Let Us Reason directly with this um we never have to sort of leave the diagrams once we've defined everything so what do we have well okay the diagrammatic equation tells us of what painting shoes does is okay well this is going to be then the same as well I empty the bucket of paint I paint this blue and you'll note that you know it doesn't matter whether I stretch things it doesn't matter if I stretch things or I twist the wires around in a symmetrical category it you know everything up to stretching doesn't matter all refers to the same configuration of states and applied processes uh but then I run into a problem here because okay well so in a symmetric normal category everything is

okay up to twist so this thing over here that crosses over I can sort of pull it I can yank it all the way to here and then what do I have well what is this saying I've just done these diagrammatic derivations to show that well I I apply the equation that tells me that I can use a full can of paint to paint a shoe and then I run out of paint and now I'm trying to apply an empty can of paint to a shoe that needs to be painted now this one I'm stuck I have no more diagrammatic equations to help me you know get this into a simplified State from here because well we might want to define the painting shoe process to be one such that nothing happens if you put in the empty you put in the empty bucket of paint don't end up painting the shoe okay for this reason not Constructors yet let us imagine then that we had this magic bucket of pain now this is I've cast a spell on it this is a magic bucket of infinite paint and the nature of this infinite paint is that it's always going to be full it's always going to have enough paint to paint another shoe of course such a shoe with such a such a bucket of paint cannot exist in our physical reality you guys have to use up some paint but if you had this sort of mathematically ideal infinite bucket of paint then it doesn't matter how many shoes you try to paint you could always then sort of apply another equation and then you go like well you know the empty bucket of the paint is not going to run out so I can I can apply the equation again and I can paint another shoe and I can keep painting shoes and every time I apply the equation well it never goes from a full state to an empty state it just stays in the same infinite state and it allows me to perform this task again and again and again however many times I want rather counter-intuitively this is what is meant by possible so painting a shoe as I've described it using just a regular bucket of paint that empties is actually not a possible task in the framework of constructive theory right this is what I mean by herbicides you know you'd want to say of course you can paint a shoe no no no no possibility is a completely political term that refers to the ability of a task to be iterated infinitely and then this thing this infinite bucket of paint here this state that enables the task the environmental state that enables the task to be repeated infinitely which is to say after you apply the task once the resulting transformation on the environment leaves it in such a state that it can do the task again nothing is a Constructor I'm gonna move on to what's next uh what is next what is next but this is a good time to ask questions this is a very good time to ask questions do you want to ask a question no I don't have any questions well why well I suppose uh what else can I say about Constructors here well conceptually where do instructors come from if you think about uh you know catalysts or some biological reactions the catalyst is the the notion of a catalyst in chemistry and biochemistry this is this is one of the sort of yeah the sort of intellectual sources

of this idea of a Constructor you know if you replace Constructor with Catalyst it doesn't really change much about how you should Intuit the thing it's this idea that well you know you need it in order to help this help this uh transformation occur but then at the end you just get the Catalyst back and then the catalyst is unchanged and you can go and participate in another participate in another reaction right so so that's this is the sort of idea of like what a Constructor is um okay so why again is uh what should I talk about here we can talk about okay we still have to get to this idea of the cut so how is how is it that this how is it that these uh having this notion now of a Constructor helps us Define what we mean by a cut yeah that's it cut if possible and not possible okay so what is a Constructor let's I'm going to rehearse one more time this uh I'm going to simplify it uh what I want to do here yeah okay and I have now this this infinite bucket of paint and I have all the shoes I'll get to in a second for whatever reason constructive theorists really like to use the category of sets and relations they really like to think of things in terms of sets and relations which is a problem when you're trying to model Quantum things and we can get into that in a second um how did this come about well you know the usual way which is to say when David Deutsch wrote his first paper he said well you know let's just you know let's take certain relations as this sort of setting where we can start doing some calculations and we'll use it as a placeholder setting for now and of course you know no one no one bothers to ever go back and change the placeholder than it then the placeholder becomes a standard so okay fine they use they use sets and relations and then so the the systems in the process Theory we'd call them systems but in the uh constructive theory we would call them substrates or the environments no there's going to be sets they're going to be sets of going to be sets of and again we're dying under the terminology here sets of macro States finally we want let's just talk about macro States well okay I mean like this is a this is a deep Constructor Theory lore thing because they need to just they need to talk about like thermodynamics in a second so why macro States because there's lots of different ways a shoe can be blue you know if you're really looking at it at the at the molecular level you know there's all these different configurations and you know you kind of don't care and you want to say like well look all of them just count as blue as far as I'm concerned we'll call all of that we'll call all of that the the macro States of the the shoe being blue and then there's shoe being whatever other color and you don't really care and then the job of this task drop of this task here is just to take the macro states that are not blue and to turn them into macro states that are blue and it must be so for well it doesn't matter if this is a set that's going to be a set of microstates macro States turns out doesn't matter when you're working process theoretically

we show in the paper that like look there is this uh there is this distinction at the mathematical level when constructed theorists talk about this between like a particular molecular configuration and the macro states of a collection a sub-collection of configurations that are blue or whatever but you know what we show is like look look in this system it doesn't matter you can just work directly with the macro States because that would that's what you really want and it turns out mathematically it doesn't make a difference so what are we saying a Constructor is here's a definition of a Constructor we're saying that this equation all to hold right what I mean by this I mean by this fill in the language of uh in the diagrammatic language for sets and relations you've got a special State and you've got a special something that's the co-state or an effect something that only takes inputs and doesn't give you any outputs what's this special state remember states in the symmetric model category of relations satellite relations are subsets and this special state is the special subset of everything in the so what this part the diagram says here just up to here saying okay well look here's my Constructor that's in my environment and my substrate is well okay I don't care what you put in here it could be any macro state but you perform this task and then the opposite of give me everything well here is the effect the relation that is the converse of this this uh well something that takes every uh it's something that takes every element of a set to the signal Singleton so it's it's the equivalent of deleting or forgetting about it and in total what this is saying here look if you apply this if you apply this process view it as a task it takes this Constructor in the environment acting on a substrate now look if you if you forget about it marginalize over or you just Trace out or whatever words you want to use to say just forget about it it doesn't matter what you put in in the substrate you're going to get back something that is good to go again as a Constructor and now rather than equality because of working in sets and relations and recall that these states are now subsets we'd say well look if I give you if I give you a Constructor what I what I mean to give you here is a sub I'm going to give you a a set or a collection of macro states that all work as Constructors um because of course the infinite bucket of paint might also have Micro States you know the paint could be in different configurations but they might all be equally infinite and good for the pro good for this task but I give you all of these and if you apply the if you apply the process to it and you do the task you get back you get back a set of states that is good to go again as uh to to run this you can just plug that into another copy of this equation and you can just get it again and again this is now minus the now we can just take away all of the sort of stuff that's still tethering us to intuition and at that point that is just the definition of a constructive right this is that that's a

construction so what can we do uh what can we do with uh Constructors so Recall now this is this is going to be the Constructor and this is by virtue a task which by virtue of having a Constructor that's going to be a possible task and now final bit that we need to get to to end the this conceptual story of what's going on in the paper and we can move on to a discussion is uh well what it what do we mean by this cut remember there was this cut between you know so those all these conceivable tasks and then there was a cut then you had all these possible tasks and then all these impossible tasks how do you get that cut again you know you have a sub-process theory so process Theory process theory is closed under composition in two ways you can compose processes and sequence and you can compose processes in parallel process Theory says well you give me a bunch of processes and the processors well I'll give you the closure all the conceivability is the closure of these processes under sequential and parallel composition and just you know what I mean by a sub-process theory well in the same way that you can have a subgroup or you can have a uh you can have a vector space that embeds into another you know a sub something of some other mathematical construct we're saying well you can also have another process Theory where you start off with a subset of processes and then you take the closure of that under sequential and parallel and twists in the case of the symmetric model category and then you end up with well that's also a process Theory and it can be a process Theory that's a subset of a larger process Theory and so translating all of that into so so translating the sort of mathematics there into sort of touching the Constructor Theory we'd be saying that look what we would have to show in order to claim that you know this definition is somehow good for where constructive theory uh construct a theorists want to go is we want to say that the possible tasks themselves form the possible tasks you know what do I mean by that well that means just that you know if I give you or identify where the possible tasks are which tasks are in an ambient process Theory then these possible tasks with form a sub symmetric minimal category which is to say the sequential or parallel composition of possible tasks again yields a possible and that is not too difficult to show that is not too difficult to show and perhaps like you can you know this is the sort of thing that you can this is the sort of thing that you can try at home if you have if you have the pen and paper follow along all I have to do well once you have this equation I mean all you have to do is check you know because this is this sorry this in equation this substitute relation must be verified to define whether a task is possible this this inequality simultaneously defines possible tasks and Constructors so uh that has to be verified for every sequential composite that you can come up with and every parallel composite that you can come up with and that's not so bad I'll show

you show you one maybe um again questions welcome is the mind a Constructor is the Mind constructed yeah well that depends on your depends on what you mean it depends on depends on the the the process Theory or this or whatever other mathematical Incarnation you pick for the constructive theory remember it's a meta Theory you bring your own mathematics so I can't answer that question until the mathematics is brought and they're you know all the English has been turned into math and then I can say oh yes that's a short or no so it depends that's pretty good I like that okay what is the okay here let's say I have you have two tasks I'm not a fan of using letters I'm more of a fan of using colors but if anyone is uh if anyone is colorblind or if anything's not clearer then just let me know maybe there's a different yep that's then just to make sure everything matches up to this end so here is how here is how we're going to sequentially compose tasks right uh this here is the task uh task one task two both tasks they work on the same substrate so for example paint issue and then unpaint issue sequential composition of tasks well if you're going to sequentially compose it needs to be on the same substrate otherwise sequential composition that's meaningless and you might want something like parallel composition instead so for example you can compose the tasks which may not be possible tasks so paint the shoe and then strip the shoe of paint I don't like it you know do something else with it um and here's how we'll sequentially compose just conceivable tasks if you have a conceivable task that's operating on this orange substrate you need to compose something else that takes oranges that substrate again but what about the environment the environment can differ like for example if I'm going to paint a shoe with paints versus if I'm going to I don't know do one of those elect electrolysis chrome dipping of a shoe to get it covered in Chrome the environment is going to be different right so for the first task I have this particular well the first task I have this particular environment the second task that has its own those environments are just systems like say that well okay if I'm going to sequentially compose I feed in the requisite environment to do task number one and then I twist in and feed in the two and then just to make sure that everything type checks at the end this entire thing here now maybe it's better if I like gather these guys here to make it clear all right that's sequential composition what does it saying right you can always gather systems can also be composite systems that are placed next to each other this entire thing says well you've got to give me the environments for the two different tasks and the substrate then I'll give you back the environments again for the two different tasks and the substrate that's how we achieve the sequential composition of tasks so these two small put them together just like this and then you get this one big task what about parallel composition parallel composition

looks really similar the parallel composition you can compose whatever doesn't really uh apparently composition doesn't particularly matter if I get this right here so we got substrate one yeah or an orange we got substrate two over here I'm gonna do it like this that makes it easier than that I got environment one and then environment two and then those are your two tasks and this is your composite test is that all visible from there I wonder hopefully that's kind of a difference yeah fantastic so again what are we saying well we've got if you want a parallel compose parallel composition of task is fantastic you don't need to match up with the substrate you can simultaneously on one side paint shoes and the other one go you know do your taxes or whatever it is is another task whatever so you can have two completely different substrates doesn't particularly matter you have two different environments for those substrates in that particular task and here's the parallel composition of conceivable tasks you know same principle if you understand sequential composition and uh you're happy it's just intuiting the twisting wires then you're good and now to verify that these guys are so now that to verify that the composition of possible tasks is again a possible task s now let's say that these guys are possible tasks in other words that there is a particular instructor or green and there's a particular Constructor for yellow over here and remember to check the condition and just delete those right so how do we go about verifying this what we want to do is we want to apply the screen one here first we probably want some kind of formal condition on our tasks on our possible tasks to tell us that this goes through to then get us that we have this series of inequalities I'm winging it here but there's going to be a proper worked out version you know the longer version of this through that goes in I'm happy to take questions as well and longest over here messing around and then apply it again I think in the paper we do like one of these Renovations it's so hard the principle is kind of simple you apply the equation that's uh sorry the end the the containment relationship that defines possibility on uh on both of these one by one and you also make use of the fact that well there's a special state that one that corresponds to everything and delete pull those through as well for tasks then you end up with this and then you go okay fine I verified that this is uh verified that this is a possible task and then you do the same for parallel composition and then you're done then you say okay well if I know that if I sequential and parallel composition Plus checking the twists okay that then I have another process Theory which means that my possible tasks form a a process Theory so what do I have I get to say I get to finally give mathematically formal voice so what a constructive theorist might want to do so again I have this first I mean here what is that that is going to be this space of conceivable tasks that's going to be some symmetric monitorial category just telling

me this is the domain in which I'm interpreting a particular process Theory and these in this symmetric nodal category well my conceivable tasks they can compose and sequence and in parallel what I mean by the cuts or the cuts that distinguishes what is possible for impossible well it turns out that the possible stuff that is also a symmetric monoidal category and that's a symmetric group minority category that embeds into the larger one it embeds in the same way that a subgroup embeds it embeds in the same way that um a sub a vector Subspace or linear Subspace is going to embed into a larger and then everything else well then everything else is going to be well you know no promises this is not the sort of stuff this is you know impossible stuff is the stuff that doesn't necessarily iterate instantly because they don't have Constructors and if you followed along so far and there's more details in the paper for if you want to like look up formal details uh you understand now Constructor Theory minus principle of locality which perhaps not interesting enough to Merit uh further comment but um if someone's curious about it we can talk about it uh but you know so what I presented so far um is you know a constructive theorists would also be happy to say yes this is constructive theory minus principle of locality so you don't go around you know saying that this thing is just constructive theory to a constructive theorist because they'll get mad at you and they'll say no no it's missing the principle of locality but that's a dogma and you don't you don't particularly need it like that's it messes a lot of things up okay so that's that's all I have to say for now I think and uh yeah happy to take it to a discussion now awesome wow um Muhammad would you like to give a first comment or reflection or just share what what angle you participated in the work from yeah I am interested in applying this formalism to uh the various examples that constructive theory has been applied to so you might have seen papers by constructed theory of information and constructive theory of life and I'm also interested in seeing how it applies to quantum theory and if the process view can be shown to be uh superior to the constructed theoretic field cool so if you have a given system of Interest like let's just say a cognitive system like a person or an aunt or something like that like how do you go about demarcating what tasks are in play are these cognitive or are these the tasks that the body performs how do we begin to to analyze a situation in a way that's amenable to to Being Framed like this so but like this could you could you refine the uh the discourse preference what do you mean by this yeah if if we want to understand if we want to understand a given informal actual embodied system in a way that's amenable to all of the compositional Bountiful benefits what are we actually looking to break down about that system of Interest sure [Music] when we talked about so tasks are kind of kind of this ad hoc thing right is remember that when we do tasks

we drew them like directed in this way because we had to say something about production and then we distinguished between their substrates and their environments and then we said that uh systems are always these things which sort of have there you're sorry the task is something that takes in the system and the environment and returns to the same system and environment which is which is something that you you can just accept but also you know in general a process doesn't need to have the same inputs and output types so this is this is perfectly fine as well you can have a process that's hit like that and maybe for some sorts of you know at the whatever level of abstraction you care about um it might be more well there's maybe there's level abstraction where you don't care about the actual physics of what's going on where everything is sort of like conservative on some level you might just care about a view of the system where you have inputs and output types in which case the language of or the the framework of thinking about tasks is uh perhaps less fundamental I mean one of the points that we're making here is like the framework of thinking and task is a task is a constructed concept on top of the more fundamentally primitive notion of just being able to construct processes right and and or just being able to compose processes so then it would depend now on on what you sort of want from um your notion of task so what what we were able to show is that look just using the language of process Theory we were able to code and formalize this particular notion of what we want tasks to behave like but then depending on what you would consider a task to be maybe the same maybe different um you can encode it in this in this mathematical language and then start reasoning about it in a diagrammatic way and there's a lot of things you can reason about in process theoretic there's a there's a thought there's a laundry listed button on the paper like electrical circuit Theory quantum theory most notably um the neural algebra first order logic yeah it's all good maybe that's maybe that's not quite a satisfactory direct answer but perhaps we can refine the question and the target so to add to add to it uh it's up to it's up to the modeler or the user to the site which processes are to be modeled so you can I mean if you want to if you want to think about cognitive processes then you are you know you can theater sub-symmetrical category to work with that but you can also have a more mechanical process theories or you can even have one where both mechanical and cognitive processes interact so it's up to the model I would say there's no hard and fast rule in the formalism it's the shoe painting Factory um but a lot a lot to say on that um inactive inference we're often interested in a general cybernetic entity or thing something that's engaging in sense making on the inbound and on action selection decision making on the outbound and another important idea from ecological psychology is this notion of like extended cognition and the

environment and the interactions of the agent environment so I wondered is the substrate you were twisting different environments in and keeping the spotlight on the thing it is that how you track an entity as things change around it is by twisting in and out different contexts or um so that's that is just the way for that is just a way to define a it's a fine to Define uh the sequential composition or the parallel composition of tasks in a way that relies upon a more primitive notion of the sequential and parallel composition of processes the twisting in and keeping a spotlight on a particular context so so one of the things that so one of the things that we've seen um that you we've all seen in this in these examples is that uh you can start with a process theory of like basic processes that just compose in sequence and in parallel and then you can define a new way to compose in sequence and in parallel on top of and that would give you a new symmetric model category that would give you a new so you can cut yeah this is so this is a process Theory kind of there's a dog food thing here somewhere where you're kind of eating your own dog food right um in the case of the the twisting which was what was the twisting again here's the twist thing we were saying that okay well look we're going to Define instead of just plugging things together we're going to define a new kind of uh we're going to find a new kind of sequential composition that looks like this where instead of just taking two boxes and joining them together we do all this other we do all this other business over here like this is this is this other business over here allowed and good and expressive but that's also something that the mathematics has no stance on like if you so you you brought in these Concepts like this is how you're spotlighting a particular substrate and keep and switching out the context right and if you look at the diagram it's perfectly okay to start pointing at these things saying well of course you're spotlighting the substrate and you're switching out the contexts but that's on you that's on you the mathematics just just tells you like this is a way to compose in sequence um but it does so nicely enough in a way you know where where this sorts of uh the sort of we can we can bring our linguistic intuitions to bear on the situation in a much more direct way because we have the visuality of it available to us that's very deep the necessary and sufficient conveyance is on the Blackboard and then that enables us to take perspectives and direct attention and have procedural flows with our attention and tell narratives and and again have interpretations but the interpretation isn't what's written on the screen yes all right a question from the live chat can you please explain in what way does that cut I believe between the possible and impossible allow the modeler to express immersion phenomena or laws that arise from the underlying processes in Constructor Theory important processes and laws that arise from the underlying in what ways does the cut

allow the modeler to express emergent phenomena or laws that arise from the underlying yeah well um if I could just ask for clarification from live chat here and maybe happens I can weigh in as well but like what's your well what's a what's your favorite toy example of the emergence I'm not a constructive theorist but you know we can play with the system and we can play with the system live and see what we we get at but first we're gonna you know what are you thinking in terms of immersions people mean different things sometimes when they say that right about the the arising directionality of a flock of birds the arising directionality of a flock of birds oh as in there's a dynamic system that's evolving over time which is you've got these birds that are interacting with each other locally and then they and then they end up with these so I think the safe answer or my intuition would say that the vast majority of things you would consider to be possible are not actually technically possible in Constructor Theory right so right off the bat you can say look you've got this intuition so so if you thought painting shoes was a fine sensible thing and you think that birds flying around and interacting with each other is a fine sensible thing that's possible that probably isn't what is meant by possible so so in the case of like a in the case of like a bird system you've got this massive interacting system fine a Constructor a cons you know you probably could have like a sort of infinite shotgun Constructor that just stops all the birds dead in their tracks again and again and you can you can kill as many birds as you want a little construct it for this sort of system be well if you can set up a if you can set up an environment that sort of steers the birds a possible task maybe for a chaotic system the only possible task is the trivial one where you don't expect anything to be done and you accept any sort of outcome uh but if you want the particular possible task that's tantamount to programming like a flock of birds or finding some way to program a thought flock of birds like a deterministic and set way that you can keep doing it again and again right so so what is the relationship between Constructor Theory and programming um there's that's a pretty good question um I'm going to try my best to answer on behalf of constructive theorists here um but there there's this uh there's a sort of there's an information Theory or a person or or perspective on information theory that also comes about from constructive theory so as a as a math and computer scientist myself I'd say that perhaps the most fundamental uh point of contact between constructive theory and programming is in a notion of information and what in particular you're allowed to do with it so in a Quantum setting you cannot clone information your information is carried in your qubits and you have no cloning so you can't you can't clone information um whereas and everything's not everything that's not classical you know the technical terms say well everything's not Quantum you call it classical in the

classical setting you can copy information you can write down a one or a zero on a piece of paper and you can get something to copy them but one and zero as many times as you want you can stick the ones and zeros into filing cabinets so and by so doing perform computation with copying let's look at the difference there is well okay if you've got different sort of underlying capacities for copying things and not being able to copy things in your underlying process Theory I guess the sorts of there's questions asked about the sorts of programs you can write but that's kind of kind of a at a much more fundamental level I mean Constructor Theory if you if you gave me a if you gave a particular process theory of programs then we can start talking constructive theory um Constructor theorists so I think by and large are more interested in the physics which is at a much lower and more basic level of abstraction so I I don't know when you're so when you're asking about like is this you know is this a get rich quick scheme when it comes to emergence is this a get rich quick scheme when it comes to you know programming and understanding chaotic Behavior etc etc you know sorry but the answer is no like there's there's nothing the extent I mean if you can unders you understood the content of constructive theory in about like an hour and it was all the pictures like there's nothing more complicated than that everything else beyond that is you hallucinating interpretations onto the pictures it's a get epistemically rich quick scheme good episode yeah yeah yeah yeah get epistemically rich quick you know you've got the pictures and they're very suggestive but you know in principle it's no different from like uh a tarot reading you've got just a very suggestive of mathematics and and you can read whatever meanings that in you want and you can play with the mathematics you can also get some meanings out but you know you only get it out as much as you put in in terms of like uh epistemically weight-bearing interpretations of hallucinations in the first place right well it's an interesting date with Constructor Theory and process Theory and active inference none of which are opinionated in The Last Mile about any given system but all of which are providing us higher levels of expressivity and increasingly inactive inference category theory is being used to represent the kinds of generative models that are used and so there's the math and the analytical equations and then there's the kinds of Bayesian graphs that we've traditionally looked at the partially observable Markov decision processes and then in the last years there's some string diagrams and all these other features coming in that are helping like expand the scope of What kinds of cognitive models and the relationships between substrates and cognitive processes that we want to explore indeed I I I'm familiar with the friends with Toby Toby Sinclair Smythe and uh yeah that's that's my only contact to active inference but I can I can definitely see the um the appeal of having a the the

appeal of having non-opinionated mathematics for for things that in the last mile as you say don't have an opinion and but but you know I'm saying I'm butchering it I'm tired and I'm saying it all wrong but I'm saying that look I think it's a good idea to have like a map of composition to deal with a complex system that's fundamentally compositional um so what do we what modules do we load in or what do we bring in in our to be able to to load up and actually use so he said bring your own math that was right at the beginning so it's like okay what do I bring pianos axioms or what do I bring like my f of x or what I mean by the mathematics that has to be brought so when it comes so so the the sort of a basic uh the the the core module that you sort of get with process theories is a theory of composition that works for uh anything that you can name processes that act on systems and these processes can compose in sequence and in parallel this is the this is the sort of basic conceptual map that you get and becomes gun to head formal when you load it into a process Theory that's that's what you get that's core everything else uh is actually expressed like sort of in terms of the in terms of the diagram so for example this piano's axioms right let's talk about the what is it nice I forget what the axioms are but I know that there are two generators you've got a zero you got a zero State and then you've got a successor box right and maybe the first thing that you want to do if you want to actually well you want to load in um anything that you can express at the end of the day in terms of algebra and symbols and equations is also doable with also doable like diagrammatically like for example I believe that in the piano formulation the addition operation is something that's sort of uh that behaves like a homomorphism with respect to or like a module with respect to the successor box so something like something like this is right the axioms are proto-procedural in that they Define exactly what can and can't be done indeed indeed so the the sort of benefits of thinking in a or let's say depicting in a process theoretic way is that well look if you're more used to talking about processes and how they interact and you you don't really care you don't want to you know spell out from the bottom up how they're implemented and do a sort of calculation with like the specifics what you really just care about or the operational constraints of how they compose and behave when you compose them together and you can express them in terms of equations or containment relations or whatever else then the process theory is much better as opposed to for example just dealings with the sets you know the you know especially if you've got like a very complex system then you know no hope again to these two key features of cognitive systems like the sense making on the inbound and the the associated World model and then the action selection I've I thought about the first blob you drew with the conceivable that's like well you drew the lines we have

the rest of the Blackboard so it's like we know that there's unconceivable but that's that's the scope of the sense making for a given setting and then the the line between what is possible and what isn't possible is is what is actionable I guess you can develop thought settings where okay well I'm gonna um speculate what could happen if I could lift up ten thousand pounds and then you could do tasks related to moving cargo containers whereas if you um were using tweezers and you defined what was possible with with a different threshold then there'd be different kinds of plausible tasks and yeah so yeah so for the sort of what what might interest you is well I'm going to Shield it now why not I've got some time my interest to you for this sort of reading is that well for Constructor Theory you know it really just wants to force you on one side of a binary and this may be too restrictive ex you know in terms of expressive capacity when you're trying to especially map it onto something that involves anything as complex as the distinction between cognitive and physical tasks and processes right and and we had questions to this uh to this effect earlier as well like what do we do with the likely what do we do with the vanishingly unlikely what do we do with the plausible right how do we so I want to point out that and this is something that's coming in the longer version of the paper is uh is a relationship between um uh constructive Constructor Theory as a special case of resource theories so you know this idea that you have to with the constructive theory you have to either go like infinitely many times or you know infinite or bust a resource theory is sort of going like yeah but you know we'd be happy also if we could do it a couple of times you know tell me how many times you can do it you know that's pretty good and so you know resource Theory and principle give you this idea of like well you know here's the sort of things that you can do once you can do twice that you can do three times you can do and finally you know in the in the very very core over here you know that's the sort of stuff that's picked out that's the sort of stuff that can be picked out by a constructive theoretic Cuts in the limit whereas a resource theoretic point of view is saying like well of course you know you if you want to make more distinctions than that then like an easy generalization that's also quite simple to think about is let's talk about the stuff that we can do for free all the time let's talk about the things that we only you know we only get to do once um and it turns out that resource theories are particularly you know resource theories are actually like a practical bit of mathematics when it comes to uh Quantum circuit optimization because there are certain components that are very expensive and you want to be able to work out what sort of quantum maps you can implement only using this particular sat gate let's say like three times or maybe you get like one extra one or maybe you get like two less and then you know the resource

theory is going to tell you precisely you know what your gain and loss and expressive capacity is when you have more few fewer or more resources which is something that is a very computer sciencey down to earth practical thing as opposed to just the distinction between infinity and everything else The Constructor Theory gives you but sneak peek for for someone that's coming the next to the next installment or the extended version of this paper that's going on archive that is super exciting and relevant certainly organizations or any other kind of system that's being designed and having an ability to handle the finite um very relevant here's one other kind of speculative question so of course we had the ice cream belting and unmelting yeah yeah yeah yeah you had the ice cream um on melting as impossible yeah but of course yeah yeah in a freezer in a freezer it does happen and or at room temperature with a reverse time Arrow and so yeah within this notion of the um Gibbs free energy we have catalysts like protein enzymes which are not able to enable what isn't possible but rather they can accelerate by reducing the activation energy and they can accelerate a negative Delta G and they can accelerate something that is possible catalytically hence Catalyst and be regenerated right and it made me think about the change in variational free energy as the ball rolling downhill cognitively and about how the the channels and the Landscapes of the mind or of the cognitive system um sometimes the landscape is changed and what is possible becomes different but just because something's possible doesn't mean it's ever reached and then we still need this question of the catalyst to actually Implement and get it to where it can be right right and and I I think that perhaps one of one of the sort of intuitive appeals of Constructor theory is that you know at first glance and with an intuitive mapping of Concepts this sound Constructor Theory sounds like the right kind of mathematics to talk about you know these these interactive Concepts that you're talking about the free energy and catalysis as enabling but the caveat with constructive theory remember is infinite robust hard binaries which means that if you want to do so like I'm sorry to say but if you want to do anything that's remotely nuanced more nuanced than like a pair of qubits interacting which already because of principal locality gives Constructor theorists a lot of pain because soccer theory is probably not the right system for you process theories though if you want to deal with composition not bad especially you know you gotta you get diagrams for free and also you get other add-ons that are very good at doing the sort of things that you want to do let's have the right expressivity and shape like resource theories Constructor Theory one should think of as a particular way or methodology uh that's designed for a particular problem to be a meta Theory of physics in in whatever David in whatever David Deutsch considers to be like a sort of

conceptually clean setting of just being able to make a single cut to tell you about like to distinguish in a black and white way all of all the physics interesting and in the Bayesian mechanics which is some of the recent developments in active inference and free energy principle area that yeah grounding of to I guess what we want to call physical even though we know that so many asterisks have to be added when we knock on the wood and everything like that but if the physics of cognitive systems can be described purely informationally then we can have a procedural compositional cognitive physics that would enable us to do anything from the mind conceptualizing that the shoe could be a different color to the multi-agent setting and composed across individuals and there's just there's there's more walls to paint on well there are many possible walls to paint on um yeah I'm what are you curious about her I mean what what's an exciting Direction in terms of uh in terms of what exactly because in terms of uh this whole business with constructive theory this is kind of a side project for me a fun one but just the side interest um what's your what's your main interest or what what motivates or what excites the context such that this could be a side interest well uh you know my main interest is more uh I suppose it's more aligned with this notion of sense making um my my interest is sort of using diagrams not only for sense making but as a formal uh platform to you know articulate theories of what since making one might be so I played around with so these diagrams that you know they're good for Quantum for example but the same kind of mathematics is quite good for other sorts of positive you know there are other domains not just for example psychology that are you know clearly in need of some sort of compositionally native approach like diagrams like Linguistics for example um so I like to draw language and because language is a particular tool of sense making and then to formalize the the kind of sense making that we make for example when we no we use we use metaphor to like to get the point across relies on a complex metaphor where information is stuck into words as a box and then the boxes moved across a conduit and this sort of sense making um is beyond the merely let's say truth theoretic uh yet nevertheless is systematic in a way such that the compositional aspects of the language precisely map onto the compositional aspects of the domain of sense making um and I suspect that's where we might have more to talk about um because you know sense making is I consider sense making it's a it's you know capital S very it's a it's a very big thing and perhaps like the important activity to and I'm interested in the systematicity of it and in particular trying to depict that's systematicity that's something yeah there's something that's you know direct about the about string diagrams in the same way that perfume or music is sort of directed it's not mediated by you know by the by the rows of Greek and the symbols right

so I don't know maybe maybe that's uh wow there's a lot there can we say that the the screen is the substrate that we're writing language on if we're going by hand with pen and then we have an infinite pen and this is this is the inexhaustible that certainly does seem to be one thing at which the yeah yeah yeah I mean you mean to refer to in the context of the mind we have a I mean as long as we're not a fantastic we have a sort of theater of mind that's sort of infinitely refreshable paper or infinitely refreshable uh stage for us to arrange things on and and and sort of replenish the representation that does sound kind of like a if we're if we're happy also to sort of sort of rule out heraclitus or heraclitus you mentioned hurricane was earlier you know a man never steps to the same river twice you know are we insofar as we are the same entity that is perhaps observing the theater of Mind as we're infinitely refreshing it then fine yeah it's a Constructor but is it the same if if you know 10 years on you're you're a functionally different person is that still uh is it is it still the same task is it still the same task refreshing or looking at a memory 10 years on I don't know the I is able to to Nest sub tasks however it may not have the awareness to know what it is embedded within but at least within this internal theater it can make a sub representation that it believes could be wrong but that it believes would have continuity also Beyond itself but also not saying what is beyond itself Beyond itself in in what sense like there's the environments that one is confronted with and then one could at least suspect that there are other environments that one is not being confronted with but also know that they can't say anything more than that about environments that they aren't being confronted with yes I I'm not sure what to say to that but I'll have to think about that once before we have to go um well I'm I'm very good um just I'd like to everyone to know that the longer version of this very paper is uh going to come out soon ish on archive where uh we work out some more examples we relate to constructive theory to Resource theories and everything is going to be very readable so if this condensed Journal version is a bit Technical and difficult to follow watch the space and there's going to be a uh longer and more pedestrian version that hopefully everyone can follow yeah great thank you both for joining hope to um see you again it's a very exciting and evocative line of work thank you thanks Daniel hope to see you again soon farewell bye bye